ENGO 531/632
Advanced Photogrammetric and Ranging Techniques

Photogrammetry Lecture Assignment

Date given: 8 October 2026

Date due: See schedule below

Marks: This assignment constitutes 40% of your final grade.

Objective
The objective of this assignment is to create a lecture on an approved topic about methodology related to ENGO 531/632 that is not covered in the lectures and labs. You are to prepare a video presentation of your lecture for asynchronous delivery. You are also responsible for the partial assessment of the video produced by another group/student.

For students in ENGO 531
This is a group assignment. You may be in the same group as for the lab assignments.

For students in ENGO 632
This is an individual assignment.

Proposal
You may choose your own topic. However, topics must be approved by the course instructor. Each group/student is to submit a proposal having the following structure:

- Lecture title
- Name of group (keep it tasteful)
- List of group members (ENGO 531 only)
- Required contents
    - What is the subject of your lecture?
    - Why is it important to geomatics engineering? Why is it important to society in general?
    - Topics that will be covered in your lecture
    - List the topics to be researched and presented by each group member (ENGO 531)
- Source material
    - List the references you will draw upon in your research on your chosen topic. Avoid Wikipedia and grey literature.
    - Provide at least two references per group member (ENGO 531) or five per student (ENGO 632)

- Provide complete and accurate bibliographic details of the references—the choice of bibliographic style to use is yours, just make sure it is consistent. The Student Success Centre provides resources for many commonly used citation styles:

https://www.ucalgary.ca/student-services/student-success/writing-support

The list of topics in the background may be preliminary—this is fine. It should not simply be a listing of generic headings such as "Introduction, Theory, Application and Conclusion" but an initial plan of what you plan to do. The length of each submission should be one page maximum, excluding references. Topic approval will be communicated within one week of submission. Note that revisions may be requested if multiple groups choose the same topic.

Video lecture

Your task is to prepare a video lecture on your chosen topic for asynchronous delivery. This is to be an introductory lecture. The target audience is ENGO 531/632 students. The audience will not become experts but will be suitably informed about the topic after watching your video. Your lecture should present the theory of the methodology supported with properly referenced examples. It should not just be an overview of an application of photogrammetry.

The duration of your video should be between 30 and 40 minutes, depending on group (ENGO 531) size. (In a real classroom setting, this would allow time for discussion and questions.) Each ENGO 531 student should plan to speak for 10-12 minutes. For ENGO 632 students, the duration should be about 30 minutes. The exact time is not as critical as how you make use of the time. Ensure that you give clear explanation of the material instead of just quickly reading the slides. A short presentation in which the speakers race through the material with shallow description will not score well.

It is recognized that not everyone will have access to the same technology to produce video content. Creation of a professionally edited video recording is neither expected nor required. PowerPoint has audio and video recording capabilities as well as graphical tools and animations to allow you to prepare a professional presentation. An audio recording over PowerPoint slides is acceptable. Your video should be uploaded to YouTube or Vimeo. It should be public or unlisted, not private. The link is to be provided in a transmittal letter submitted to Dropbox.

Possible topics

The field of photogrammetry is vast, so there are many possibilities for suitable topics. Some potential topics include, but are not limited to, the following:

- High-definition maps
- Photogrammetry for digital construction
- Deep learning
- NERF
- Gaussian splatting
- Mobile mapping
- Panoramic imaging
- Forestry
- Image matching
- Structure from motion (SfM)
- Underwater photogrammetry

- X-ray photogrammetry
- Point cloud segmentation
- Time-of-flight (ToF) and RGB-D sensors
- Simultaneous localization and mapping (SLAM)
- Full-waveform laser scanning

This project will require independent research of books and scientific literature. A good starting point is the International Society for Photogrammetry and Remote Sensing (ISPRS). All the proceedings of ISPRS conferences are freely available here:

https://www.isprs.org/publications/archives.aspx

and here

https://www.isprs.org/publications/annals.aspx

The official journal of the ISPRS

https://www.sciencedirect.com/journal/isprs-journal-of-photogrammetry-and-remote-sensing

can be accessed via the library here

https://ucalgary-primo.hosted.exlibrisgroup.com/primo-explore/jsearch?vid=UCALGARY&lang=en_US

Other relevant journals include the Photogrammetric Record

https://onlinelibrary.wiley.com/journal/14779730

and Photogrammetric Engineering & Remote Sensing

https://www.asprs.org/asprs-publications/pers

You can discuss potential topics with the instructor at any time.

Assessment and due dates
Each presentation will be assessed by the instructor and by your classmates. Anonymous peer review is an indispensable part of scientific publishing, grant applications and the evaluation of professional activities for licensure, so it is likely that you will encounter it in one form or another during your career, if you have not already.

Each student must watch and assess one presentation of another group. The instructor will assign the group that you will assess. You will be provided with the link to the video and an evaluation rubric. Your assessment will comprise numerical scores and written comments. The scores will contribute to the final score achieved by the group you evaluate. Your identity as a reviewer will remain anonymous.

The submission of the peer review assessments is mandatory for all students. Peer reviews should be unbiased, fair and constructive. Your peer assessment will be graded by the instructor. The instructor reserves the right to exclude a peer assessment that is inappropriate (malicious or vacuous). If a peer

review assessment has not been submitted or has been excluded, it will be replaced by the instructor's assessment.

The due dates, submission method and penalties for each component of this assignment are listed below. Note the harsh late penalties due to strict deadlines for grade submission. Extensions will not be granted except under extenuating circumstances.

Component	**Submission method**	**Due date and time**	**Late penalty**
Group members and topic outline	PDF of proposal to Dropbox	Sunday 25 October 2026 at 23.59	50% per day late for this component of the assignment for the group
Video posted to YouTube or Vimeo	PDF transmittal letter with video link to Dropbox	Friday 4 December 2026 at 23.59	50% per day late for this component of the assignment for the group
Completed peer review forms	PDF of review form to Dropbox	Friday 11 December 2026 at 23.59	50% per day late for the assignment for the individual

The distribution of marks for this assignment is given below. Note that whilst this is a group assignment, there is an individual component based on the instructor's assessment of your peer reviews.

Component	**Type of assessment (ENGO 531)**	**Marks**
Topic outline and group members	Group	5
Video assessment by instructor	Group	20
Video assessment by student peers	Group	10
Peer review assessment by instructor	Individual	5
Total		**40**

Note that all assessments are individual for ENGO 632.

Questions?
Please feel free to contact the instructor if you have any questions.

END OF ASSIGNMENT